

Power BI Basics

Question 1: What is Power BI and why is it used in businesses?

Power BI is a business intelligence and visualization tool by Microsoft. It helps organizations connect to multiple data sources, transform raw data, and create interactive dashboards. Businesses use it to make data-driven decisions, identify trends, and share insights efficiently.

Question 2: Name and explain the three main components of Power BI

1. **Power BI Desktop** – For creating reports and dashboards.
2. **Power BI Service (Cloud)** – Online platform to publish and share reports.
3. **Power BI Mobile App** – Mobile access to reports and dashboards.

Question 3: Explain the Power BI workflow

1. Connect to data sources.
2. Clean and transform data using Power Query.
3. Model data and create relationships.
4. Build visualizations and dashboards.
5. Publish and share reports via Power BI Service.

Question 4: List any four data cleaning tasks in Power Query

- Remove duplicates.
- Handle missing values.
- Split or merge columns.
- Change data types.

Question 5: Step-by-step instructions to load the Real_Estate dataset

1. Open **Power BI Desktop**.
2. Click **Get Data** → Select source (Excel/CSV).
3. Browse and select the **Real_Estate Dataset**.
4. Click **Load** to import.
5. Use **Power Query Editor** for cleaning if needed.

Question 6: Define Data View, Report View, and Model View

- **Data View** – Shows raw data tables.
- **Report View** – Used to design dashboards and visuals.
- **Model View** – Displays relationships between tables.

Question 7: Different data sources supported by Power BI

- Files: Excel, CSV, XML, JSON.
- Databases: SQL Server, Oracle, MySQL.
- Cloud services: Azure, Salesforce, Google Analytics.

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- APIs and online services.
- Microsoft services: SharePoint, OneDrive.

Question 8: Split Owner Name into First Name and Last Name

1. Open dataset in **Power Query Editor**.
2. Select **Owner Name** column.
3. Use **Split Column** → **By Delimiter (space)**.
4. Rename new columns as **First Name** and **Last Name**.